



Birla Institute of Technology & Science, Pilani

Pilani Campus

ACADEMIC— UNDER/POST GRADUATE STUDIES DIVISION

First Semester 2026-27

Course Handout Part II

Date: 4th August 2026

In addition to part I (General Handout for all courses appended to the Timetable), this portion gives further specific details regarding the course.

Course Number : CE G551

Course Title : DYNAMICS OF STRUCTURES

Instructor-in-charge : DIPENDU BHUNIA

[Team of Instructors] : NONE

- 1. Course description:** Introduction and Scope of dynamic analysis of structures; origins of vibration theory and experiment; review of earlier concepts: D'Alembert's principle, equations of motion. Elements of a structural system: springs, dashpot, mass; Springs in parallel and series; methods to formulate equations of motion; Formulation (equation of motion) and solution of undamped and damped free vibration analysis of S.D.O.F system; Formulation (equation of motion) and solution of undamped and damped forced vibration analysis of S.D.O.F system; Forced vibration under harmonic, periodic, impulse, step, ramp, general dynamic forces (time and frequency domain analysis) and response spectrum load, support excited vibration, seismic pickups; Numerical techniques for evaluation of dynamic response of SDOF system; time domain analysis; direct integration techniques; finite difference method (Central Difference method); Newmark beta method; average and linear acceleration method; Development of equation of motion and solution for two degree of freedom systems; Free vibration analysis of MDOF systems; frequencies; mode shapes and response; orthogonality condition of mode shapes; Approximate methods for obtaining natural frequencies and mode shapes; Holzer method; Stodola's method; Rayleigh's method; Rayleigh-Ritz method; Inverse iteration method; Vector iteration method; Rayleigh's Quotient iteration method; Matrix iteration method; Generation of damping matrix for MDOF; dynamic properties; modal damping; classical damping; damped response with Rayleigh and Caughey damping. Mode superposition method; mode acceleration method; modal combination rules using absolute sum, SRSS and CQC method.

2. Scope and Objective:

Natural phenomena and human activities impose forces of time-dependent variability on various civil engineering structures. This course deals with analysis and design of structures subjected to dynamic loads, which involve consideration of time-dependent forces. Based on this scope, the following objectives have been listed:

- To make this course necessary for students who are desirous of joining design offices/ industry related to buildings, industrial plants, bridges, tanks, offshore structure, tall chimneys etc.
- To determine the behavior of structures under dynamic loading and design a structure which is resistant to dynamic loading such as seismic, wind, seawave, vehicle induced vibration force etc.

3. Course learning outcomes:

- a. Development of software for the analysis of framed/skeleton structures.
- b. Capability development for the use of structural dynamics concept and its applications for framed structural analysis and design under dynamic loading.
- c. Determine the behavior of structures under dynamic loading and design a structure, which is resistant to dynamic loading such as seismic, wind, seawave, vehicle induced vibration force etc.

4. Text books and references:

TB1. Anil K. Chopra, "Dynamics of Structures: Theory and applications to earthquake engineering", Prentice Hall India Ltd., 2007.

TB2. Mukhopadhyay, M., "Structural Dynamics: Vibrations & Systems" Ane Books Pvt. Ltd., 2006.

RB1. R.W. Clough and J. Penzien, "Dynamics of Structures", McGraw Hill International edition, 1993.

RB2. Paz, M., "Structural Dynamics: Theory & Computation" CBS Publishers & Distributors, 2001.

5. [Method of conduct of the course]

The course will be conducted primarily through Chalkboard with ppt as well online medias such as Google class and/or Nalanda.



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6. Course Plan: weekly schedule for 14 weeks with [3] lectures per week

Week No	Module	Lec No.	Reference	Learning Outcome
1	Introduction and Scope of dynamic analysis of structures	1-2	TB1-1, RB1,RB2	Origins of vibration theory and experiment; review of earlier concepts: D'Alembert's principle, equations of motion.
1 - 2	Elements of a structural system	3-5	TB1, TB2, RB1,RB2	Springs, dashpot, mass; Springs in parallel and series; methods to formulate equations of motion
3-4	Free vibration analysis of S.D.O.F system	6-9	TB1, TB2, RB1, RB2, Class notes & Assignments	Formulation (equation of motion) and solution of undamped and damped free vibration
4-5	Forced vibration analysis of S.D.O.F system	10-13	TB1, TB2, RB1, RB2, Class notes & Assignments	Formulation (equation of motion) and solution of undamped and damped
5-6	Forced vibration under various dynamic loadings	14-17	TB1, TB2, RB1, RB2, Class notes & Assignments	Harmonic, periodic, impulse, step, ramp, general dynamic forces (time and frequency domain analysis); response spectrum load, support excited vibration, seismic pickups.

6-7	Numerical techniques for evaluation of dynamic response of SDOF system	18-21	TB1, TB2, RB1, RB2, Class notes & Assignments	Time domain analysis; direct integration techniques; finite difference method (Central Difference method); Newmark beta method; average and linear acceleration method
7-8	Two-degree-of-freedom systems	22-25	TB1, TB2, RB1, RB2, Class notes & Assignments	Development of the equation of motion and its solution
9-10	Free vibration analysis of MDOF systems	26-29	TB1, TB2, RB1, RB2, Class notes & Assignments	Frequencies; mode shapes and response; orthogonality condition of mode shapes
10-12	Approximate methods for obtaining natural frequencies and mode shapes	30-35	TB1, TB2, RB1, RB2, Class notes & Assignments	Natural frequencies and mode shapes
12-14	Generation of damping matrix for MDOF	36-42	TB1, TB2, RB1, RB2, Class notes & Assignments	Dynamic properties; modal damping; classical damping; damped response with Rayleigh and Caughey damping. Mode superposition method; modal combination rules using absolute sum, SRSS and CQC method. Response Spectrum.

7. Evaluation Scheme:

(Given below are suggestions for possible kinds of evaluation components.
Indicative only.)



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Component	Weightage	Duration	Date and time	Comments
[Course Project]	[25%]	[50 min]	Throughout the semester	One seminar before midsem, i.e. 01/10/26 in 1204 and another seminar before compre but after midsem, i.e. 27/11/26 in 1204
[Homework Assignments]	[20%]	[one alternative week]	Throughout the semester	[Each assignment after completion of each module; however not more than 6 assignments in total and 1 will be buffer]
[Mid Semester]	[25%]	90 min	10/10 AN2	[Close Book]
Comprehensive	[30%]	180 min	16/12 AN	[Open Book]

Criterion for NC: If the student gets less than 20% of the average class marks, he/she will be awarded NC.

2. **Chamber consultation hours:** Friday from 4 to 5 PM or based on the appointment by email.

8. **Makeup Policy:** Makeup for only midterm and comprehensive examinations will be given provided the prior request is made with genuine reason.

[Dipendu Bhunia]
Instructor-in-charge
[CE G551]